

Battery Storage In Westchester: A Simple Fact Sheet



What battery energy storage is, how it performs, and how it's regulated in New York.

This fact sheet is a plain-language reference on the technology of BESS, its local track record, and the safety standards that govern it.

1 WHAT IS BATTERY ENERGY STORAGE (BESS)?

A battery energy storage system saves electricity for when it's needed most: it charges when power is cheap and abundant, and delivers it on hot summer afternoons and evenings when the grid is under the most stress. These are not e-bike batteries: utility systems must pass safety certification, permitting, code review, inspections, and local emergency planning before they operate.



Small (Tier 1)

10–80 kWh

Homes and small businesses. Footprint: a small refrigerator.



Medium

under 600 kWh

Commercial backup power and demand management. Footprint: one parking space.



Large (Tier 2)

over 600 kWh

Grid support and community energy storage. Footprint: about half an acre.

2 WHY IT MATTERS: RELIABILITY, BILLS, AND AIR

Keeps the lights on

Demand is rising fast as homes, vehicles, and heating electrify. Storage delivers power during the highest-demand hours, cutting overload and outage risk in constrained parts of the grid.

Holds bills down

The grid is built for a few peak hours a year. Storage charges in the valleys and discharges at the peaks, deferring wire and substation upgrades that ratepayers ultimately fund. NYC customers paid **\$4.5 billion** over a decade keeping 16 old fossil “peaker” plants on standby.

Cleans the air

Fossil peakers fire up on the hottest days, exactly when air quality is already worst. Storage reduces how often they need to run.

3 SOLAR NEEDS STORAGE

Solar + storage work together.

Paired with solar, storage saves locally produced clean power for when it's most useful.

Renewables are intermittent.

The sun sets and the wind calms. Batteries turn intermittent generation into dependable power.

Solar + storage = affordability.

Solar saved New Yorkers about **\$90 million** in peak-hour costs last summer alone. Storage makes that clean power available at the evening peak.



Croton-Harmon Station: solar, BESS, and EV charging in Westchester

IT'S ALREADY WORKING HERE



White Plains Community Solar

6.8MW solar + 7,056kWh storage. **\$1M annual revenue** via a lease-based model with NYPA.



IBM Watson Research Center

5.5MW carport, 6.5M kWh/yr, 85 EV chargers. Subscribed employees save **~10% a month**.



Vail GMC, Bedford Hills microgrid

75% peak demand reduction, 84% cost savings. Monthly bill fell from **\$49K to \$7,840**.



Croton-Harmon Train Station

4.2MW canopies + 3.5 MW BESS + 24 EV chargers. **\$480K annual lease**, offsets 400 homes.



GHP Office Realty, Harrison

375kW / 940kWh battery earning **\$75,000 in annual revenue** on shared savings, zero upfront.



Regeneron, Tarrytown

550kW solar carport deploying clean energy onsite while supporting high-demand operations.

Demand response works here too: Sustainable Westchester GridRewards has ~1,000 MW of peak reduction pledged and 100,000+ Con Edison customers enrolled. The average residential user earned \$80; New Rochelle earned over \$100,000.

5 SAFETY: PROVEN TECHNOLOGY, LAYERED PROTECTIONS

- ✓ **Battery management systems** keep cells in safe temperature ranges and can catch a fault 50–80 charge cycles before it becomes a problem.
- ✓ **Smoke, heat, and gas detection** with automatic shutoff and 24/7 remote monitoring.
- ✓ **Steel fire-rated enclosures**, firewalls, and spacing between units.
- ✓ **Emergency procedures pre-planned** with local fire departments, plus BESS-specific responder training and on-call experts.
- ✓ **Certified before deployment**, UL 9540 equipment certification, UL 9540A fire testing, NFPA 855 installation standards, and New York's updated fire and building codes.

All energy involves risk; the question is how well it's managed. Every year, on average, the U.S. experiences:

- **125,000** gas leaks (22,000 in NYC alone) • **286** oil pipeline incidents
- **27,900** home heating fires • **167,800** home cooking fires

Together: 254 deaths, 3,084 injuries, and \$2.5B+ in damage from fossil-fuel systems, every year.

Two New York incidents, compared:



2014: East Harlem gas explosion

Leaking gas pipeline. Two buildings destroyed instantly. 8 deaths, 50+ injuries.



2023: Warwick, NY BESS incident

Manufacturing defect in two battery units. Fire contained on site, no injuries, and air monitoring showed toxin levels below harmful thresholds.

Importantly: the battery type and format of the Warwick facility is no longer the protocol for modern BESS. Under the state BESS standards already in place today, **this failure could not happen now.**

6 THE FUTURE OF ENERGY IN WESTCHESTER

New York is the epicenter of BESS restrictions, holding more than 80% of every active moratorium in the country, and Westchester is among the most restricted counties. The proven alternative is a tiered local law that permits storage under clear safety and siting standards your board controls. Adopting it as a starting point means a town regulates storage responsibly instead of prohibiting it, and CEAC can help customize the language for local conditions:

Equipment & safety testing	Site layout & setbacks
Emergency response plans	Fencing & landscaping
Noise & visual screening	Stormwater management
Operating conditions	Decommissioning plans

Working examples nearby.

Croton-on-Hudson, a village within the Town of Cortlandt, adopted a tiered battery code in 2024, and projects reviewed under it are operating today. Elmsford and Tarrytown have working BESS codes as well.

What a model law looks like: two tiers set by capacity (in Croton, Tier 1 up to 200 kWh and Tier 2 above that); small systems approved with a building permit, larger systems through Planning Board review, with certified equipment (UL 9540), fire-department coordination, setbacks and screening, and a decommissioning plan, following the structure of NY's model law. Bedford can adapt this proven language rather than start from scratch.

HANDOUT PROVIDED BY

The BCW Clean Energy Action Coalition (CEAC): 130 businesses, nonprofits, municipalities, and institutions across Westchester, convened by the Business Council of Westchester with EarthKind Energy. CEAC offers municipalities model code language, engineering and fire-safety experts for hearings, and plain-language resources for residents, at no cost. Learn more: thebcw.org

Sources: NYSEDA Energy Storage Program & Battery Energy Storage Guidebook; NYISO 2026 Gold Book; CEAC Energy Storage Fire & Safety Factsheet; Town of Cortlandt Local Law 7-2025 (certified text); project figures from NYPA, Sustainable Westchester, and participating site owners. Prepared July 2026.

